

reactance (inter-electrode capacitance, lead inductance), transit-time effect, cathode emission, plate-heat dissipation, power loss due to skin effect, radiation and dielectric loss.

Tubes that are efficient in the microwave range usually operate on the theory of velocity modulation. The microwave tube uses transit time in the conversion of DC power into radio-frequency power. The interchange of power is obtained by using the principle of electron velocity modulation and low-loss resonant cavities in the microwave tube.

Velocity modulation is defined as the variation in the velocity of a beam of electrons caused by the alternate speeding up and slowing down of the electrons beam. This variation is usually caused by a voltage signal applied between the grids through which the beam must pass. The directions of the electron beam and the static electrical field are parallel to linear beam tubes. Against this, the fields influencing the electron beam stand vertically by the electron beam at the crossed-field tubes.

Magnetron—a microwave device

Magnetrons are a special form of diodes. Electrons move between the cathode and anode in a curved fashion, and thus the electric field and the magnetic field are normal to the electron beam. When an electron is slowed down by an electric or magnetic field, it gives up energy, making the field stronger. If an electron's speed is increased by an electric or magnetic field, it weakens the field.

An amplifier tube circuit that generates radio frequency signals is called an oscillator. A resonant circuit consisting of an inductor (coil) in parallel with a capacitor determines the frequency and wavelength of the oscillator. The lesser the number of turns in the coil, the smaller the capacitor plates and the higher the radio frequency that the oscillator generates.

Future technology

1. Multiple-beam klystrons for synthetic-aperture radars and missile seekers
2. TWTs for towed decoys
3. Microwave power modules based transmit/receive modules for phased-array radars
4. High-power microwave devices for directed-energy weapons
5. Gyro TWTs for radar applications
6. Vacuum microelectronics based microwave devices (TWT on a chip)
7. Tera-hertz devices for secure high-data-rate communication, imaging and radar
8. Microwave power beaming rectennas
9. Microwave propulsion
10. Microwave plasma chemistry
11. Microwave-generated artificial ionospheric mirrors (over-the-horizon radars and battlefield illumination)

Magnetrons have a central cylindrical cathode surrounded by an anode in the form of a thick cylindrical shell. Top and bottom plates form the remainder of the vacuum envelope. These plates are placed between the poles of a strong magnet.

The controlled resonant circuit problem at microwave frequencies is solved by using hollow metal cylinders as a resonator. The round cylinder walls are similar to a one-turn coil and the cylinder end plates are like a very small capacitor. These cylindrical resonators are called microwave cavities. By moving one of the cavity's end plates in or out of the cylinder, the frequency of the oscillator can be tuned. The cylinder must be about a wavelength in diameter and about one-half wavelength long at the resonant frequency. Energy can flow in and out of the resonator through a hole in the cylinder wall.

The microwave fields from the resonators extend into the region between the cathode and anode. A strong magnetic field makes the high-speed electrons move such that these don't reach the anode and return to the cathode, unless slowed down by giving up energy to a cavity electric field.

If the electrons arrive at the wrong time, these take energy from the microwave field, speed up and spiral back to the cathode. At the correct voltage and microwave frequency, the electrons move at the correct velocity to continue to loose speed and give up most of the

energy to the microwave field before the impact on the anode.

Other devices

Magnetron oscillator was the first device developed that was capable of generating large powers at microwave frequencies. Later, improved devices such as travelling-wave tube amplifiers (TWTAs) were developed for use in microwave systems. Yet, magnetron production continues for use in microwave ovens.

Cross-field amplifier (CFA) is another microwave power amplifier. It is a cross between TWTs and magnetrons in its operation. It has a magnetron structure to provide interaction between crossed DC electric and magnetic fields on one hand and RF field on the other. It also uses a slow-wave structure, as in TWT, to provide a continuous interaction between the electron beam and a moving RF field.

The backward-wave oscillator (BWO) is also a microwave-frequency and velocity-modulated tube that operates on the same principle as the TWT. However, a travelling wave that moves from the electron gun end of the tube towards the collector is not used in the BWO. Instead, the BWO extracts energy from the electron beam using a backward wave that travels from the collector towards the electron gun (cathode). ●

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